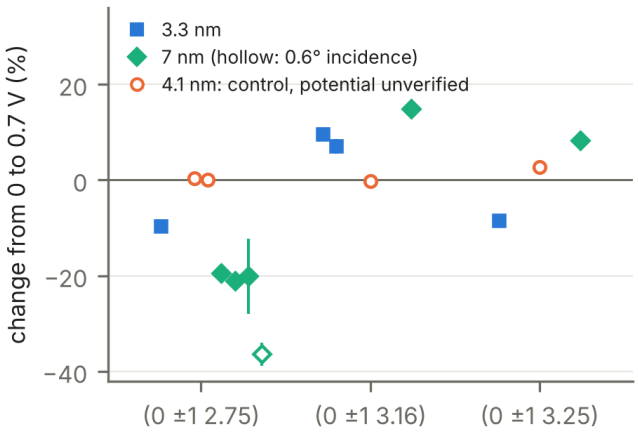
RuO_2(110) films in 0.1 M H_2SO_4: what the in situ SXRD data support, with potential control of the 4.1 nm film treated as unverified

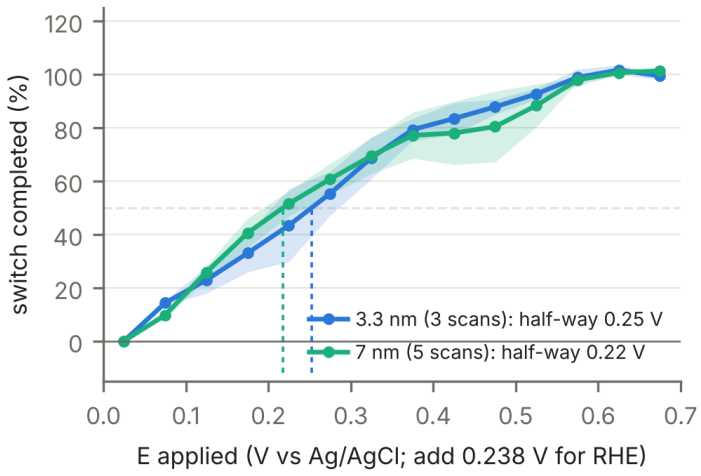
NSLS-II, October 2026. Epitaxial films on TiO_2(110), 3.3, 4.1 and 7 nm. Ag/AgCl reference; RHE = Ag/AgCl + 0.238 V (logbook). Stationary CVs at fixed reflections: 30 s at 0 V, then 0 to 0.7 V at 5 mV/s. CTR rods at fixed potentials.

- **Bottom line.** The 3.3 and 7 nm film surfaces switch reversibly between two states near 0.22-0.25 V vs Ag/AgCl (about 0.46-0.49 V RHE), at the reflections the CTR model marks as surface-sensitive, while OH-blind reflections stay flat.
- Both films switch at the same potential within the scan spread, although the 7 nm film is partly relaxed and the thinner film is expected to be fully strained.
- The 4.1 nm film is shown only as a negative control: its current record was lost, so potential control is not claimed. Under the same cell, beam, protocol and analysis it shows no signal locked to the potential.

(a) Change from 0 to 0.7 V, one point per CV scan



(b) Switch completed vs potential, both sweeps averaged



(a) Error bars: ±1 s.e., the larger of a noise-surrogate and an autocorrelation-robust error. (b) Per scan, 0% is the first bin (0-0.05 V) and 100% the plateau (0.55-0.7 V); bands: ±1 s.e. over scans.

1. The switch, without a model (3.3 and 7 nm)

Each X-ray point gets the potential EC-Lab applied at that moment (SPEC and EC-Lab clocks agree within 1-2 s); per scan, a slow drift plus the change from 0 to 0.7 V is fitted. Weighted per reflection:

reflection	3.3 nm	7 nm
(0 ±1 2.75)	−9.7 ± 1.3%	−20.3 ± 0.7%
(0 ±1 3.16)	+9.3 ± 0.5%	+14.9 ± 0.9%
(0 ±1 3.25)	−8.5 ± 0.8%	+8.3 ± 0.8%

- **Surface fingerprint:** 2.75 falls while 3.16 rises, in both films. Beam, absorption or normalization artifacts would move all reflections the same way.
- **Reversible:** anodic and cathodic sweeps overlap at 5 mV/s; the swing is the same at 25 mV/s (scan 222).
- **Reproducible:** 7 nm (0 1 2.75) measured four times (Ru spot 30, 42, 42, 42%); 3.3 nm (0 −1 3.16) twice.
- **Surface-specific:** OH-blind control reflections stay at their noise: 7 nm (2 0 1.75) 0.2% peak-to-peak; 3.3 nm (−2 0 1.75), (−2 0 1), (−1 0 2.32), (1 0 1.75) 1.1-3.5% (point noise 2.4-4.0%).

2. The 7 nm CTR rods follow the potential

- Whole (0 1) rod, rms of ln(ratio) with one free scale: 0.7 vs −0.1 V gives 0.077 and 0.080 (both directions), against 0.039-0.040 for repeats at the same potential hours apart.
- Surface states fitted to the CTRs, under two bridging-O assumptions (one from Rao et al., 2017), predict the sign and onset of the 7 nm CV changes with nothing adjusted. At 0.15 V: 2.75 predicted −7 to −11%, measured −9.4%; 3.16 predicted +7 to +10%, measured +6.5%. At 0.35 V, 3.16 matches (+10 to +15% vs +13.8%); 2.75 is underpredicted (−10 to −15% vs −24.5%). 3.25 and the size at 0.7 V are not reproduced.

3. Same switch potential in two strain states

- Half-way potential: 0.25 V (3.3 nm; scans 0.20-0.31) and 0.22 V (7 nm; scans 0.17-0.24, one noisy scan aside) vs Ag/AgCl. Equal within the scan spread.
- 7 nm is partly relaxed (CTR out-of-plane strain about 0.5%); the thinner films are strained (4.1 nm CTR 1.4%; 3.3 nm not measured here). On Jeong et al.'s films (different samples), the work function differs by about 0.3-0.4 eV over this thickness range.
- **Statement:** in these two films the switch potential does not follow the strain state: they differ by 0.03 V, within the scan spread.

4. The 4.1 nm film: a negative control

- **Potential control unverified.** Every day-4 EC file shows zero current: charge at potential steps 0.003 ± 0.026 µF/V, against 28.5 µF/V on 7 nm. The cause was not found and the setup cannot be re-examined.
- Same cell type, beam, scan protocol and analysis: no signal locked to the potential in any X-ray channel (all detector ROIs and SPEC counters, judged against the beam monitors), and the (0 −1) CTR rods at six potentials agree to the repeat level (rms 0.031-0.043).
- So the 3.3 and 7 nm switch is not produced by the cell, the beam or the method.
- If the potential did reach the 4.1 nm film, it did not switch: at most 3% and 12% of the 3.3 nm response and 1% and 8% of the 7 nm response at L 2.75 and 3.16 (2σ). Show this only as an observation, not as a film difference.

5. Stated limits (not claimed)

- Absolute OH coverage: the OH fraction and the bridging-O height trade off in the fits; the relative result is the same under both assumptions.
- A thickness or strain trend: one sample per thickness.
- A mechanism, and the 3.3 vs 7 nm sign difference at L 3.25 (CV 125 was measured off the rod maximum).
- A steady anodic current on days 2-3 (µA to tens of µA) is not produced by the beam at the moment (unchanged at 3.6% transmission); its source is unknown. iR at the switch potential is under 1 mV.

6. Suggested talk (8 slides)

1. Question and answer up front.
2. Films, cell, potential program, measurement order.
3. Why (0 ±1) reflections: the 2.75 / 3.16 fingerprint.
4. Raw traces: 7 nm next to the 4.1 nm control.
5. Response at the applied potential, with controls (panel a).
6. Switch curves: the same potential in both films (panel b).
7. 7 nm CTRs and the CTR-to-CV prediction.
8. Summary, limits, and what would make it a paper: a theory estimate of the strain shift, lab CVs, second samples.

Data and scripts: analysis/2026-11-06_stationary_cv in the ruo2-ctr-sim repository (sync_abs.py, curves_sync.py, timeline_rods.py, ctr2cv.py, deep41.py, deep41b.py; this page: make_onepager.py).